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Antibody persistence to diphtheria toxoid, tetanus toxoid, *Bordetella pertussis* antigens, and *Haemophilus influenzae* type b following primary and first booster with pentavalent versus hexavalent vaccines

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ABSTRACT

Thailand has incorporated the whole-cell (wP) pertussis vaccine into the expanded program on immunization since 1977 and has offered the acellular pertussis (aP) vaccine as an optional vaccine for infants since 2001. We followed healthy children from a clinical trial (ClinicalTrials.gov NCT02408926) in which children were randomly assigned to receive either pentavalent (DTwP-HB-Hib) or hexavalent (DTaP-IPV-HB-Hib) vaccines for their primary series (administered at 2, 4, and 6 months) and first booster vaccination (18 months). Both groups received Tdap-IPV as a second booster at the age of 4 y. Blood samples were collected for evaluation of antibody persistence to diphtheria toxoid (DT), tetanus toxoid (TT), and *Bordetella pertussis* (*B. pertussis*) between 2 and 6 y of age annually, and for the immunogenicity study of Tdap-IPV at 1 month after the second booster. Antibody persistence to *Haemophilus influenzae* type b (Hib) was followed until 3 y of age. A total of 105 hexavalent-vaccinated children and 91 pentavalent-vaccinated children completed this study. Both pentavalent and hexavalent groups demonstrated increased antibody levels against DT, TT, and *B. pertussis* antigens following the second booster with Tdap-IPV. All children achieved a seroprotective concentration for anti-DT and anti-TT IgG at 1 month post booster. The hexavalent group possessed significantly higher anti-pertactin IgG (adjusted $p = .023$), whereas the pentavalent group possessed significantly higher anti-pertussis toxin IgG (adjusted $p < .001$) after the second booster. Despite declining levels post-second booster, a greater number of children sustained protective levels of anti-DT and anti-TT IgG compared to those after the first booster.

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Introduction

The diphtheria toxoid (DT), tetanus toxoid (TT), and whole-cell pertussis (inactivated *Bordetella pertussis*) vaccine (DTwP) was developed in 1931 and continues to be a fundamental element of infant immunization programs in many countries.[1] Due to its reactogenicity, the whole-cell pertussis component of DTwP has been replaced by a less reactogenic acellular pertussis (aP).[2] Both DTwP and DTaP are now widely used in combination with hepatitis B (HB), *Haemophilus influenzae* type b (Hib), and/or inactivated polio vaccine (IPV), making the combinations known as pentavalent vaccine (DTwP-HB-Hib) and hexavalent vaccine (DTaP-IPV-HB-Hib), respectively.

Despite significant reductions in disease burden due to the introduction of pertussis vaccines in the 1940s and the change to acellular vaccine to improve acceptance and coverage in the 1970s, epidemics of pertussis persist.[3] Several countries reported a resurgence of pertussis, with mortality and morbidity in infants too young to receive vaccination.[4,5] The resurgence of pertussis has been attributed to several factors, including improved awareness, surveillance, and diagnostics, new *B. pertussis* strains that escape the host- and vaccine-induced immunity, and the waning of vaccine-induced immunity, in which faster waning has been found in aP-primed than in wP-primed individuals.[6,7] To combat pertussis, it will be necessary to improve or adjust the composition and schedule of the vaccine, among other strategies.

In Thailand, two doses of the DTwP vaccine were implemented in the Expanded Program on Immunization (EPI) in 1977, which was then increased to three doses in 1982 and four doses in 1987.[8] Five years later, a total of five doses of DTwP vaccine were recommended. Primary vaccination in Thailand is scheduled at 2, 4, and 6 months of age, while the first and second boosters are scheduled at 18 and 48 months of age, respectively. For more than 10 y, Thai infants have been vaccinated with either the EPI vaccine (wP-containing) or the optional vaccine (aP-containing) according to the recommended schedule. A previous study in Thai children that focused only on the optional aP-containing vaccine showed persistence of the antibody at 4–6 y of age in individuals previously vaccinated with DTaP-IPV-Hib at 2, 4, 6 and 18 months.[9,10]

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Initial priming with aP- or wP-containing vaccine has been shown to influence the humoral immune response following a booster.[11,12] Plasma cell responses to an aP booster were stronger in individuals primed with wP than with aP vaccines.[13] Since Thailand is a country where both aP and wP vaccines are used in infants, it is possible to compare the longitudinal antibody kinetics after aP and wP vaccination. We conducted a follow-up study on children from the initial trial[14] in which infants born to mothers who received tetanus-diphtheria-acellular pertussis (Tdap) vaccine during pregnancy as recommended by the Advisory Committee on Immunization Practices (ACIP) in 2018[15] were randomized to receive either aP-containing (hexavalent DTaP–IPV-Hib–HB) or wP-containing (pentavalent DTwP–HB-Hib) vaccines at 2, 4, 6, and 18 months of age.[14] The aims of the present study are 1) to compare the immunogenicity of Tdap-IPV as a second booster at 4 y of age between children vaccinated with hexavalent (aP) and pentavalent (wP) vaccines and 2) to evaluate the antibody persistence against *B. pertussis* antigens, as well as other vaccine antigens such as anti-DT and anti-TT, up to the age of 6 y. Since Hib is an important etiologic agent of pneumonia, meningitis, and invasive infection in children aged <5 y,[16] immunity against Hib was followed until 3 y of age. To the best of our knowledge, this is the first study to compare the long-term antibody persistence induced by aP-containing and wP-containing vaccines in children born to mothers who received Tdap during pregnancy.

Material and methods

Study design

This study was an expansion study providing long-term follow-up of children who participated in a previous trial (ClinicalTrials.gov NCT02408926)[14] as shown in Table S1. In the previous trial, 315 infants born to mothers who had received Tdap during pregnancy were randomized to receive either aP-containing vaccine (Infanrix hexa, hereafter referred to as the aP group) ($n = 158$) or wP-containing (Quinvaxem; hereafter referred to as the wP group) ($n = 157$) vaccine for primary and first booster vaccination.[14] A comparison group of 79 infants, born to mothers vaccinated with TT or dT but not Tdap, was recruited after birth from the same hospital during the same period as the aP and wP groups. This group received the wP-containing vaccine (Quinvaxem) for primary and first booster vaccination (hereafter referred to as the EPI wP group). The EPI wP group represented the immune response of children under the national vaccination policy when Tdap vaccination during pregnancy was not incorporated, and wP-containing vaccines were used for primary and booster immunization. Written informed consent was obtained from parents prior to enrollment in all groups.

All eligible children in the previous trial were contacted for participation in this study. Exclusion criteria included acquired immunodeficiency disorders, chronic illness that could interfere with the completion of the trial, receipt of immunosuppressive therapy, receipt of blood or blood products before or during the study, hypersensitivity to the vaccine used in this trial, and bleeding disorders in which intramuscular injections were contraindicated.

This long-term follow-up study was conducted between May 2017 and November 2022 at the Clinical Research Unit, Center of Excellence in Clinical Virology, Department of Pediatrics, Faculty of Medicine, Chulalongkorn University, Bangkok, Thailand. In this study, children were scheduled for blood sampling visits at 2, 3 and 4 y of age. At the age of 4 y, all groups were vaccinated with Tdap-IPV (Boostrix-polio, GSK, Rixensart, Belgium) as a second booster. Blood sampling was performed 1 month after booster vaccination and then annually until the age of 6 y. This study was approved by the Institutional Review Board of the Faculty of Medicine of Chulalongkorn University (IRB 084/60 and IRB no. 173/63, CoA 528/2020) and was registered with the Thai Clinical Trial Registry (TCTR20231017001).

Study vaccine

The components of the hexavalent aP-containing and pentavalent wP-containing vaccines used for the primary and first booster vaccination have been previously reported.[14] In short, the *B. pertussis* antigen components in the hexavalent aP-containing vaccine were 25 μg of pertussis toxin (PT), 25 μg of filamentous hemagglutinin (FHA), and 8 μg of pertactin (PRN), while the wP-containing vaccine had inactivated *B. pertussis* ≥4 IU/dose of potency. The amount of DT and hepatitis B surface antigen (HBsAg) in both vaccines were equal (30 IU of DT and 10 μg of HBsAg). The amount of TT was slightly higher in pentavalent (≥60 IU of TT) than in hexavalent vaccines (40 IU of TT). The hexavalent vaccine contained 10 μg of Hib polyribosylribitol phosphate (PRP) covalently bound to approximately 25 μg of TT, while the pentavalent vaccine contained 10 μg of Hib capsular oligosaccharide chemically conjugated to CRM197.

The second booster vaccine at 4 y of age was a reduced antigen diphtheria-tetanus-acellular pertussis-inactivated poliovirus vaccine (Tdap-IPV, Boostrix-polio). The Tdap-IPV contained ≥2 IU of DT, ≥20 IU of TT, 8 μg of inactivated PT, 8 μg of FHA, 2.5 μg of PRN, and 40, 8, 32 D-antigen unit for IPV type 1, 2, and 3, respectively, adsorbed onto 0.5 mg of aluminum. The Tdap-IPV is licensed as a preschool booster in many countries around the world from the age of four. In Thailand, Tdap-IPV was an optional vaccine at the time when this study was conducted, while DTwP and the bivalent oral polio vaccine (bOPV) were in the EPI. Comparison of antigen components in each vaccine is shown in Table 1. Children were also vaccinated with other optional vaccines irrespective of the study (e.g., hepatitis A, varicella, and influenza).

Sample collection

Venous blood samples (6 mL) from all participants were collected at the ages of 24, 36, 48 months (before the

Table 1. Comparison of antigen components in each vaccine.

	Hexavalent aP-containing vaccine (DTaP-IPV-HB-Hib) (0.5 mL/dose)	Pentavalent wP-containing vaccine (DTwP-HB-Hib) (0.5 mL/dose)[a]	Tdap-IPV (0.5 mL/dose)
Trade name and manufacturer	Infanrix hexa (GSK, Rixensart, Belgium)	Quinvaxem (Crucell-Janssen, Incheon, South Korea)	Boostrix-polio (GSK, Rixensart, Belgium)
Groups that received the vaccine	aP group as the primary series and the first booster	wP and EPI wP groups as the primary series and the first booster	All groups (aP, wP, EPI wP) as the second booster
Pertussis			
Pertussis toxin (PT)	25 μg		8 μg
Filamentous hemagglutinin (FHA)	25 μg		8 μg
Pertactin (PRN)	8 μg		2.5 μg
Inactivated *B. pertussis*		≥4 IU	
Diphtheria			
Diphtheria toxoid	≥30 IU	≥30 IU	≥2 IU
Hepatitis B			
Hepatitis B surface antigen (HBsAg)	10 μg	10 μg	N/A
Tetanus			
Tetanus toxoid	≥40 IU	≥60 IU	≥20 IU
***Haemophilus influenzae* type b (Hib)**			
Hib polyribosylribitol phosphate (PRP)[a]			
Hib capsular oligosaccharide[b]	10 μg	10 μg	N/A
Poliovirus			
Inactivated poliovirus type 1	40 D-antigen unit	N/A	40 D-antigen unit
Inactivated poliovirus type 2	8 D-antigen unit		8 D-antigen unit
Inactivated poliovirus type 3	32 D-antigen unit		32 D-antigen unit

[a]Conjugated to approximately 25 μg of tetanus toxoid (TT) as carrier protein.
[b]Conjugated to approximately 25 μg of CRM 17.
[c]N/A: not applicable.

administration of the Tdap-IPV), 49 months (28–35 d after the Tdap-IPV), 60, and 72 months. Blood samples were collected in tubes devoid of anticoagulant or preservative agents. Afterward, the samples underwent centrifugation at 1500 × g (4°C) for 10 min. The resulting sera were separated and stored at −20°C until laboratory analysis. All serum samples were coded, with participant personal identifiers replaced by the assigned codes. Vaccination and blood collection schedules by vaccination group were demonstrated in Table S1.

Laboratory tests

Anti-DT and anti-TT IgG levels were quantified using (EUROIMMUN, Lübeck, Germany) as previously described.[17] Anti-DT IgG levels were divided into the following four categories: ≤0.01 IU/mL (susceptible), >0.01–<0.1 IU/mL (partial protection), 0.1–<1.0 IU/mL (protection), and ≥1.0 IU/mL (long-term protection).[18] Anti-TT IgG levels were divided into the following four categories: ≤0.01 IU/mL (susceptible), >0.01–<0.1 IU/mL (uncertain protection), 0.1–<1.0 IU/mL (protection), and ≥1.0 IU/mL (high titers).[19]

Anti-Hib IgG levels were quantified using commercially available ELISA kits (XpressBio, Frederick, MD, USA) as previously described.[17] Anti-Hib IgG levels were classified as <0.15 μg/mL (susceptible), 0.15–1.0 μg/mL (short-term protection), and ≥1.0 μg/mL (long-term protection).[20]

Antibodies to pertussis toxin (anti-PT IgG), filamentous hemagglutinin (anti-FHA IgG), and pertactin (anti-PRN IgG) were analyzed in a commercial enzyme-linked immunosorbent assay (EUROIMMUN, Lübeck, Germany) as previously described.[14] The same type of ELISA kits was used for samples collected during both the previous trials, spanning from birth to 19 months of age, and the present study.

Statistical analysis

Antibody levels are presented as geometric mean concentrations (GMCs) with a 95% confidence interval (CI). GMCs were calculated by taking the arithmetic mean of the log-transformed values of the antibody levels and then converting it to the real value (GMC) using a table of antilogarithms. Seroprotection against diphtheria, tetanus, and Hib were displayed as number and percentages as the protective antibody concentrations have been identified for these three diseases. Comparisons of the antibody levels between groups were conducted using the Kruskal–Wallis test with Dunn's post hoc correction for multiple comparisons at each timepoint. This analysis was conducted on the log-transformed data, which were found to be non-normally distributed. The statistical analysis was conducted using IBM SPSS Statistics v23.0 (IBM Corp., Armonk, NY, USA) and GraphPad Prism v9.4.1 (GraphPad, San Diego, CA, USA).

Results

Demographic characteristics of the study participants

At 19 months of age (1 month after the first booster vaccination), 329 participants were eligible and contacted for participation in this long-term follow-up study. A total of 11, 15, and 8 participants from the aP, wP, and EPI wP groups, respectively, did not agree to participate in the long-term study. Therefore, 125 participants in the aP group,

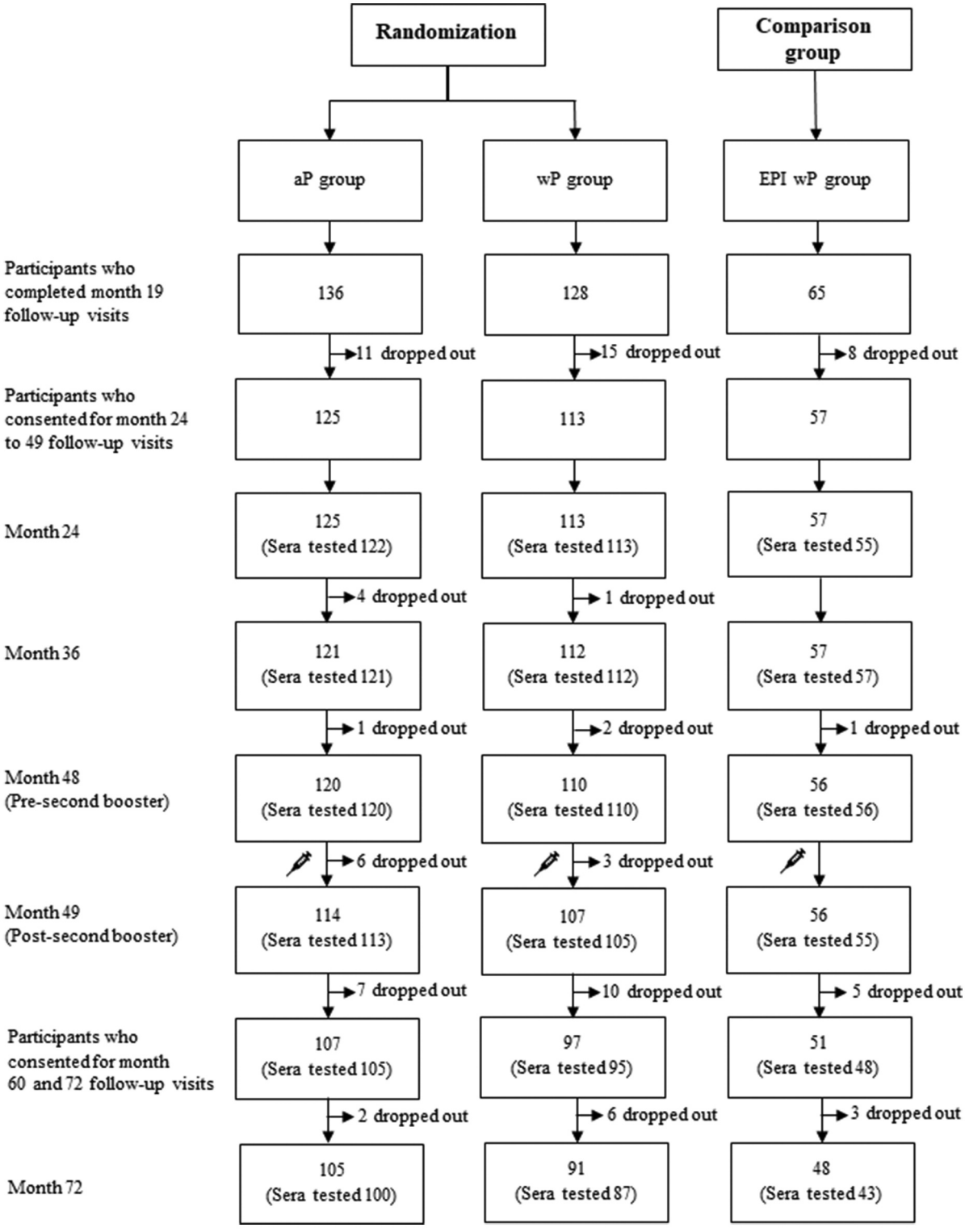


Figure 1. The consort flow diagram of this study. Abbreviations: aP group referred to participants who received acellular pertussis-containing vaccine at 2, 4, 6, 18 months of age; wP group referred to participants who received whole cell pertussis-containing vaccine at 2, 4, 6, 18 months of age; EPI, Expanded Programme on Immunization; EPI wP group referred to participants who received whole cell pertussis-containing vaccine at 2, 4, 6, 18 months of age, but they were born to women who did not receive Tdap vaccine during pregnancy. Vaccine icon denoted Tdap-IPV as a second booster vaccination.

113 participants in the wP group, and 57 participants in the EPI wP group were included (Figure 1). All had received primary and first booster vaccinations at 2, 4, 6, and 18 months with hexavalent (aP-containing) or pentavalent (wP-containing) vaccines as previously described.[14] All children received Tdap-IPV at the age of 4 y (month 48). This cohort of children was followed up until 6 y of age. A total of 105 participants in the aP group, 91 participants in the wP group, and 48 participants in the EPI wP group completed this study.

Males and females were equally distributed in the aP (48.8% male and 51.2% female) and wP groups (49.6% male and 50.4% female), while in the EPI wP group, there were slightly more male than female participants (61.4% male versus 38.6% female). There were no significant differences in mean weight and height at any of the time points studied, as shown in Table S2. The mean intervals (SD; standard deviations) between months 48 (pre-second booster) and 49 (post-second booster) were 32.1 (7.8), 31.0 (8.3), 29.7 (3.6) d in the aP, wP, and EPI wP groups, respectively.

Persistence of anti-DT IgG, anti-TT IgG, anti-Hib IgG, and B. pertussis-specific antibody at 2, 3, and 4 y of age

Figure 2 demonstrates the GMCs of anti-DT IgG, anti-TT IgG, and anti-Hib IgG of this cohort. Before the second booster, we observed a decrease in anti-DT IgG (Figure 2a) and anti-TT IgG (Figure 2b) from month 19 to month 48. Although antibody levels decreased, up to 84.2%, 71.8%, and 76.8% of participants in the aP, wP, and EPI wP group, respectively, still possessed seroprotective levels (>0.1 IU/mL) of anti-DT IgG at 48 months of age (pre-second booster) (Table 2).

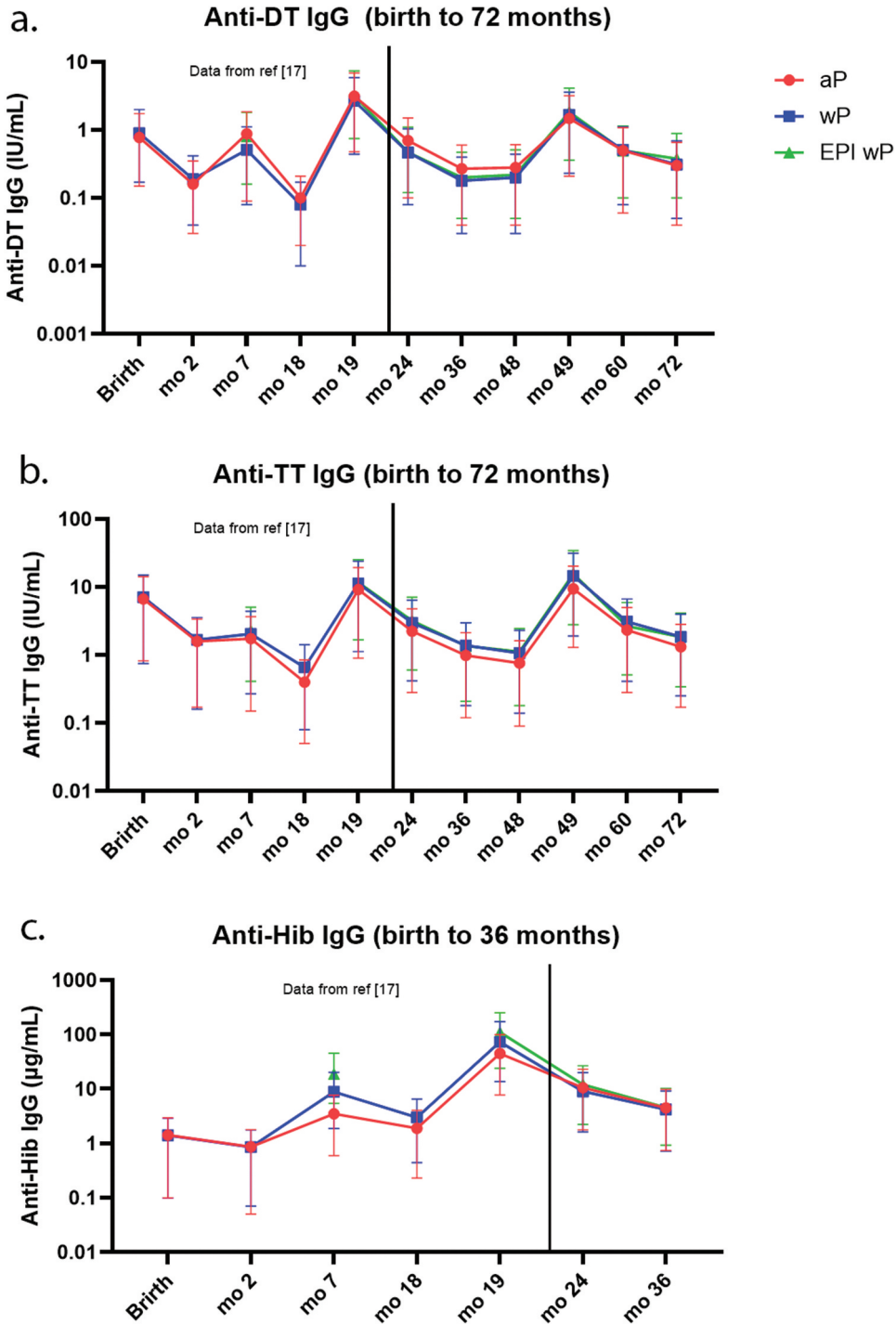


Figure 2. Geometric mean concentrations (GMCs) of anti-DT IgG (a), anti-TT IgG (b) from birth to 6 y, and anti-Hib IgG (c) from birth to 3 y among the aP (red), wP (blue), and EPI wP (green) cohort. Data from birth to month 19 were previously published.[16] Error bars denoted 95% confidence interval.

Table 2. Percentages of children achieving diphtheria, tetanus, and *Haemophilus influenzae* b seroprotection between 2 and 6 y of age.

	24 months	36 months	48 months (Pre-2nd booster)	49 months (Post-2nd booster)	60 months	72 months
Anti-DT IgG						
≥0.1 IU/mL						
Hexavalent (aP) group	95.9	74.4	84.2	100	95.2	91
Pentavalent (wP) group	84.1	56.3	71.8	100	93.7	88.5
EPI wP group	81.8	68.4	76.8	100	95.8	85.7
Anti-TT IgG						
≥0.1 IU/mL						
Hexavalent (aP) group	100	98.3	99.2	100	100	100
Pentavalent (wP) group	100	100	100	100	100	100
EPI wP group	100	100	100	100	100	100
Anti-Hib IgG						
≥0.15 µg/mL						
Hexavalent (aP) group	100	100				
Pentavalent(wP) group	100	100				
EPI wP group	100	100				

Similarly, 99.2%, 100%, and 100% of the participants in the aP, wP, and EPI wP group, respectively, still possessed seroprotective levels (>0.1 IU/mL) of anti-TT IgG at 48 months of age. Seronegative and seroprotective proportions of anti-DT IgG and anti-TT IgG from birth to 6 y and are shown in Tables S3 and S4. There were no differences in anti-DT and anti-TT IgG levels among children across all groups between month 24 to 72 as demonstrated in Figure 2(a,b) and Table S5.

Regarding anti-Hib IgG, this study followed the kinetics of anti-Hib IgG and demonstrated a decline at months 24 and 36 (Figure 2c). Despite the decline in anti-Hib IgG levels at 24 and 36 months of age, all children in the three groups maintained seroprotection against Hib (>0.15 µg/mL) at those time points (Table 2), with over 90% achieving long-term protection (>1.0 µg/mL) (Table S6). In addition, there were no differences in anti-Hib IgG levels between all groups at months 24 and 36.

Our previous study showed that at 19 months of age (1 month after the first booster), anti-PT IgG in the EPI wP group was significantly higher than in the aP and wP groups.[14] At months 24, 36 and 48, anti-PT GMC decreased over time, but the EPI wP group still possessed significantly higher anti-PT IgG levels than the aP group at month 36 (adjusted $p = .038$) and 48 (adjusted $p = .0002$) (Figure 3a, Table S7). Comparison of persistence of anti-FHA IgG between groups showed that the aP group possessed significantly higher levels of anti-FHA IgG than the wP group at month 24 (adjusted $p < .0001$) and month 36 (adjusted $p = .038$) (Figure 3b). Additionally, anti-PRN IgG was significantly higher in the aP group than in the wP and EPI wP groups at months 24 (adjusted $p < .0001$), 36 (adjusted $p < .01$) and 48 (adjusted $p < .01$) as demonstrated in Figure 3(c). There were no differences in anti-PRN IgG levels between the wP and EPI wP groups at any time points tested.

Immunogenicity of Tdap-IPV as a second booster and persistence of antibody levels up to 6 y of age

The GMCs and 95% CI for each vaccine antigen are shown in Tables S5 and S7.

All children from the aP, wP, and EPI wP groups demonstrated increased antibody levels against DT and TT following the second booster with Tdap-IPV. Additionally, all children had antibody levels above the thresholds considered seroprotective for diphtheria (≥0.1 IU/mL) and tetanus (≥0.1 IU/mL) (Table 2), with over 66% and 100% achieving long-term seroprotection (≥1.0 IU/mL) for diphtheria and tetanus, respectively. Despite the decline in anti-DT and anti-TT IgG levels following the second booster, a greater number of children maintained seroprotective levels of anti-DT and anti-TT IgG compared to those achieved after the first booster (Tables S3 and S4).

Comparison of anti-PT IgG levels after Tdap-IPV booster in the aP-vaccinated and wP-vaccinated groups showed that both the EPI wP group (wP-vaccinated infants born to mothers who did not receive Tdap during pregnancy) and the wP group (wP-vaccinated infants born to mothers who receive Tdap during pregnancy) possessed significantly higher anti-PT IgG than the aP group at month 49 (adjusted $p < .0001$), month 60 (adjusted $p < .0001$), and month 72 (adjusted $p < .0001$) (Figure 3a). There were no statistically significant differences in anti-PT IgG among the wP and EPI wP groups between month 24 to 72.

In contrast, we found no differences in anti-FHA IgG between the groups following the second booster with Tdap-IPV. For anti-PRN IgG, the aP group had a significantly higher anti-PRN IgG level following the Tdap-IPV booster than the wP group ($p < .001$) at month 49 (adjusted = 0.023) and month 60 (adjusted = 0.033) (Figure 3c).

Discussion

This study was a long-term follow-up study that compared the immunogenicity of Tdap-IPV as a second booster in four-year-old children who were primed and first-boosted with hexavalent aP-containing vaccine (DTaP–IPV-HB-Hib) or pentavalent wP-containing (DTwP-HB-Hib) vaccine. We also evaluated antibody persistence to *B. pertussis* antigens, as well as other vaccine antigens such as anti-DT and anti-TT up to the age of 6 y. Studying the persistence of antibody responses to a vaccine could lead to a plan in national vaccination strategy, as well as ensuring sustained protection throughout early childhood. Overall, we found that the majority of participants remained seroprotected against diphtheria and tetanus prior to the second booster. Additionally, Tdap-IPV, as a second booster at 4 y of age, stimulated robust humoral immune responses to all

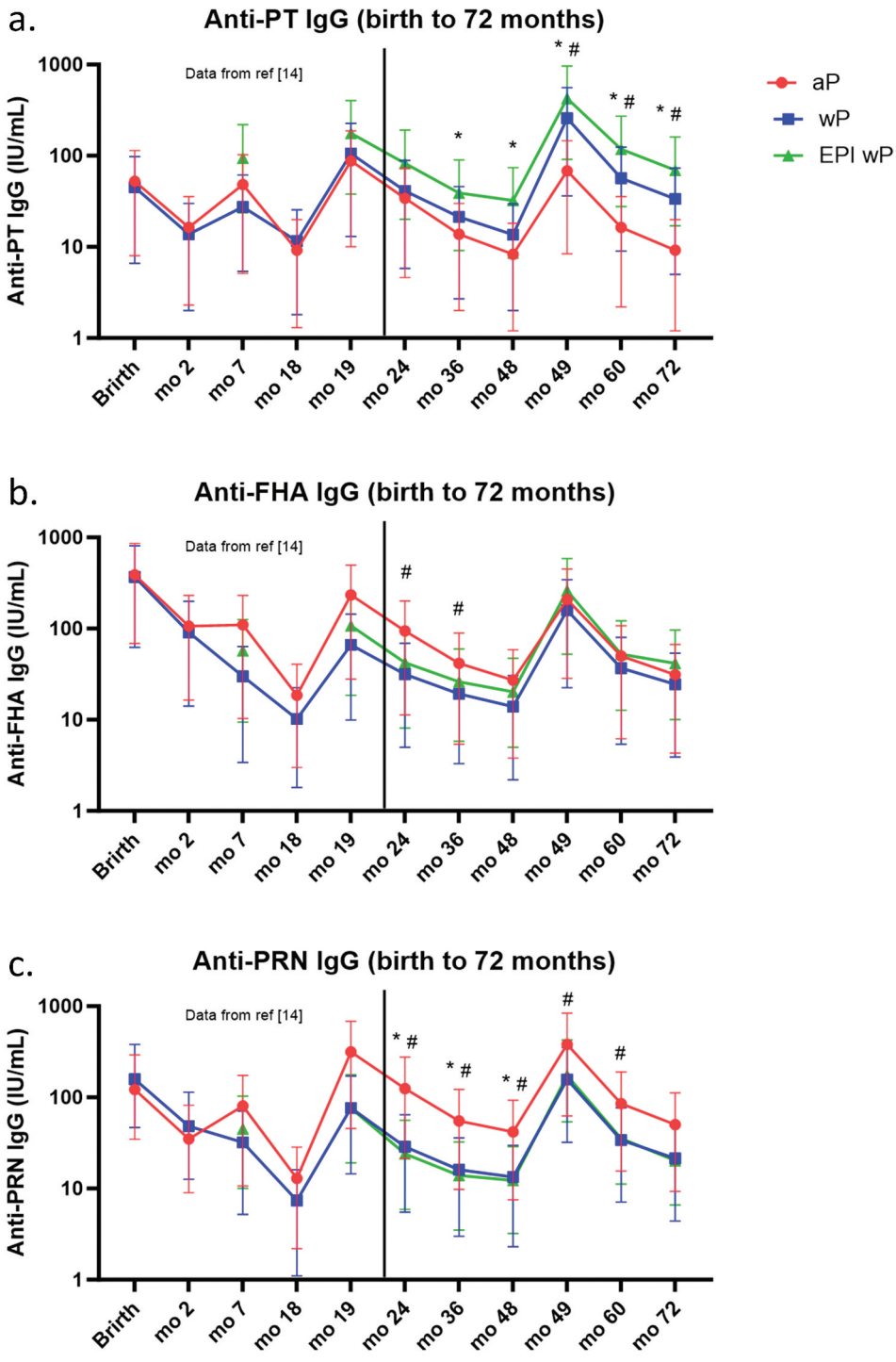


Figure 3. Geometric mean concentrations (GMCs) of anti-PT IgG (a), anti-FHA IgG (b), and anti-PRN IgG (c) from birth to 6 y among aP (red), wP (blue) and EPI wP (green) cohort. Data from birth to month 19 were previously published.[14] Error bars denoted 95% confidence interval. Statistical differences between the antibody levels among children from different groups were analyzed using Kruskal–Wallis test followed by Dunn's test for multiple comparison adjustments. Statistical significance was indicated as follows: #adjusted *p*-value <.05 between aP and wP group, *adjusted *p*-value <.05 between aP and EPI wP group.

vaccine antigens tested in this study. Even though anti-DT and anti-TT IgG levels decreased after the second booster, a larger portion of children sustained seroprotective levels of anti-DT and anti-TT IgG compared to those attained after the first booster. This finding exemplifies that administration of booster doses of DT and TT vaccines can enhance antibody levels above the protective threshold.[21]

Our study found that there was no difference in anti-DT and anti-TT IgG responses to the Tdap-IPV booster at the age of four between children primed with aP and those primed with wP. In a previous study, higher levels of anti-DT IgG were observed in cohorts primed with wP compared to those primed with aP following a second booster with DTaP at 4 y of age, while no difference in anti-TT IgG was noted.[22] In contrast, another study reported higher levels of both anti-

DT IgG and anti-TT IgG in the DTwP-primed group following the first DTaP booster.[23] Although there have been conflicting findings in anti-DT and anti-TT IgG responses at post-booster between the aP-primed and wP-primed groups, all children in the previous study[22] and the present study achieved seroprotection against diphtheria and tetanus after the second booster.

Previous studies have shown that the Tdap vaccine formulated for adolescents and adults, with reduced content of DT and *B. pertussis* antigens, could be successfully used to boost the immune responses in preschool children.[24,25] Evaluation of anti-DT IgG level after the second booster showed higher levels in the pediatric formulation DTaP-vaccinated group compared to the reduced-dose Tdap-vaccinated group.[24] However, these differences diminished over time, while both groups exhibiting comparable seroprotection rates.[24] Similarly, the GMC for anti-DT IgG after a second booster with DTaP-Hib-IPV in a previous Thai children cohort[10] was higher than with Tdap-IPV in the present cohort, probably due to the high amount of DT in DTaP-Hib-IPV (30 IU) compared to Tdap-IPV (2 IU). The modest reduction in antibody responses following the Tdap compared to the DTaP booster is unlikely to have any clinical significance, given the substantial proportion of children achieving both seroprotection and long-term seroprotection after receiving the recommended five doses of DT-containing vaccine.

A comprehensive study comparing the immune responses elicited by wP- and aP-containing vaccines in children revealed that those who received DTwP as primary series vaccination (DTwP-primed) tended to exhibit higher levels of anti-PT and lower levels of anti-FHA compared to DTaP-primed individuals after fifth dose of vaccination with a DTaP vaccine.[26] Similarly, this study found that anti-PT IgG levels after Tdap-IPV booster were higher in the wP-primed (wP and EPI wP groups) than in the aP-primed groups. The clinical significance of reduced anti-PT IgG responses remains unclear, as the correlates of protection against pertussis involve a complex interplay of immune responses to various components rather than a solitary marker.[27]

Previous reports have shown that Tdap vaccination during pregnancy is associated with decreases in humoral immune responses to *B. pertussis* antigens following primary immunization in aP-vaccinated infants,[28,29] while less consistent results were obtained following the first booster.[30,31] This phenomenon is referred to as blunting. Our earlier research[14] indicated a blunting effect on anti-PT and anti-FHA, but not on anti-PRN IgG levels, among wP-vaccinated infants whose mothers received Tdap compared to those whose mothers did not receive Tdap, following primary and booster vaccination. Regarding the longer-term effects of blunting, a study conducted in the UK showed that prior to receiving the preschool booster, GMCs of anti-PT IgG were lower in aP-vaccinated children born to mothers vaccinated with Tdap-IPV compared to those born to unvaccinated mothers.[32] However, these differences disappeared at one month after the preschool booster.[32] The present study, conducted among children vaccinated with wP, adds to the existing literature by showing that the differences in anti-PT IgG levels among wP-vaccinated infants born to mothers who received Tdap and those who did not, disappeared after the age of 2 y. Furthermore, pentavalent (wP)-vaccinated children born to women who received Tdap during pregnancy exhibited significantly higher levels of anti-PT IgG than hexavalent (aP)-vaccinated children at 1 month after the second booster up to the age of 6 y. Monitoring the kinetics of anti-PT IgG, along with other *B. pertussis*-related immune responses, will contribute to understanding the kinetics of vaccine-induced immunity and guide pertussis booster schedules in aP-primed and wP-primed children born to mothers who received Tdap during pregnancy.

Regarding seroprotection against diphtheria and tetanus following primary and booster vaccination with aP-containing vaccine, it was demonstrated in previous studies that most children still maintained seroprotection before the second booster.[10,33] The present study added that the pentavalent DTwP-HB-Hib-primed and boosted cohort also maintained a high seroprotection rate against diphtheria (71.8%) and tetanus (100%) before the second booster. Additionally, the seroprotection rate at 2 y after the second booster in our cohort remained high at >85% against diphtheria and 100% against tetanus in both pentavalent and hexavalent-vaccinated groups. Previous studies which followed children who received a reduced dose DT-containing vaccine as a preschool booster showed that seroprotection rate remained high for up to 10 y.[34] To maintain a high seroprotection rate against diphtheria, tetanus, and pertussis until adulthood, the ACIP recommended that individuals aged 11–18 y should receive a single dose of Tdap, preferably at a preventive care visit between 11 and 12 y.[15]

Our study showed that after four doses of the Hib-containing vaccine at 2, 4, 6, and 18 months of age, all children remained seroprotected against Hib (≥0.15 μg/mL) at the age of 2 and 3 y, with no significant difference in seroprotection rate between the groups. Another study in the aP-vaccinated cohort also showed similar anti-Hib persistence.[35] In terms of anti-Hib IgG levels, our previous study showed that the wP-primed cohort had significantly higher anti-Hib IgG than the aP-primed cohort after a booster dose at 18 months of age.[17] This is in agreement with a study in the United Kingdom showing that GMCs were three times higher in recipients of wP-Hib than in recipients of aP-Hib combination vaccines.[36] Differences in anti-Hib IgG between groups were likely to be of no clinical significance since all children reached the threshold for seroprotection.

This study has some limitations. First, the small sample size within the EPI wP group raises caution about the interpretation of statistical analyses. The EPI wP cohort is entirely distinct and not drawn from the same initial pool which underwent randomization. This distinction might explain the difference in antibody responses and seroprotection rates observed between the EPI wP group and those randomized to wP (wP group), despite both groups receiving the same wP-containing vaccine from the same manufacturer. Furthermore, the dropout rate from the previous randomized controlled trial might introduce some unknown biases. Lastly, wP vaccines are relatively heterogeneous due to the use of different strains of *B. pertussis* in vaccine production and variations in manufacturing methods. This complexity hinders the generalization

our findings to all wP-containing vaccines utilized in childhood immunization across different settings.[37]

In conclusion, the Tdap-IPV vaccine induced a robust antibody response in four-year-old children, with all children achieving seroprotective concentrations for anti-DT and anti-TT IgG at one-month post-booster. Anti-PT IgG levels were significantly higher in the wP-vaccinated than in the aP-vaccinated cohorts after the Tdap-IPV booster. As the correlates of protection against pertussis are a combination of immune responses to multiple components, the clinical significance of the reduced anti-PT response remains unknown. These findings enhance our comprehension on the impact of childhood vaccination across various vaccine schedules. Future studies that evaluate functional antibodies and cellular immune responses will further enhance our understanding of immune responses to childhood vaccination.

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Data availability statement

Data will be available upon request.

Authors' contribution

Nasamon Wanlapakorn: Conceptualization, Data curation, Formal analysis, Supervision, Writing-original draft, Writing, reviewing and editing, Funding acquisition. Nasiri Sarawanangkoor: Data curation, Formal analysis, Writing-original draft. Donchida Srimuan: Project administration. Thaksaporn Thatsanathorn: Project administration. Thanunrat Thongmee: Investigation, Methodology. Yong Poovorawan: Conceptualization, Data curation, Formal analysis, Funding acquisition, Supervision, Writing – reviewing, and editing.

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